

Drug Review Insights & Summarization Tool

ADS-509 Final Project — Generative AI Application

This report presents an end-to-end analysis of 219,206 patient drug reviews from the UCI ML Drug Review Dataset. Traditional data analysis and visualizations were produced using Python. Plain-language summaries were generated using the Google Gemini 2.5 Flash API. The goal is to make patient experience data accessible to pharmacists and healthcare researchers without requiring a data science background.

1. Overall Dataset Summary

AI-Generated Summary

Here is a summary report of the patient drug review dataset:

1. Dataset Overview

This dataset comprises a substantial collection of 219,206 patient drug reviews covering 4,211 unique drugs and addressing 2,724 unique conditions. The reviews have an average rating of 6.99 out of 10, with a median rating of 8.0 out of 10, indicating a tendency towards higher ratings. On average, each review is approximately 85 words long.

2. Sentiment Landscape

The overall sentiment within the dataset leans positive, with 145,106 reviews (66.2%) classified as positive. Negative reviews account for 54,474 instances (24.9%), while neutral reviews make up 19,626 instances (9.0%). This positive trend is supported by the rating distribution, where the highest volumes of reviews are observed for ratings 10 (68,973 reviews), 9 (37,321 reviews), and 8 (25,794 reviews).

3. Notable Patterns

"Birth Control" is by far the most reviewed condition, with 38,436 reviews, significantly more than "Depression" (12,164 reviews) or "Pain" (8,245 reviews). Among the top-reviewed drugs, several birth control medications like Levonorgestrel (4,930 reviews, avg rating 7.4), Etonogestrel (4,421 reviews, avg rating 5.8), and various Ethinyl estradiol combinations consistently appear. Notably, Phentermine stands out with a high average rating of 8.8 across 2,085 reviews, while some high-volume birth control drugs, such as Ethinyl estradiol / norethindrone and Nexplanon, have lower average ratings of 5.6.

4. Limitations & Caveats

Researchers should note that this data is patient-reported and structured, meaning it reflects user experiences and sentiments rather than clinical outcomes or objective medical assessments. The dataset provides summary statistics but lacks demographic information or clinical history for reviewers, which could influence reported experiences. Interpretations should also consider that the average review length is 85 words, which may offer limited depth of detail for individual experiences.

Supporting Visualizations

Figure 1: Overall Rating & Sentiment Distribution

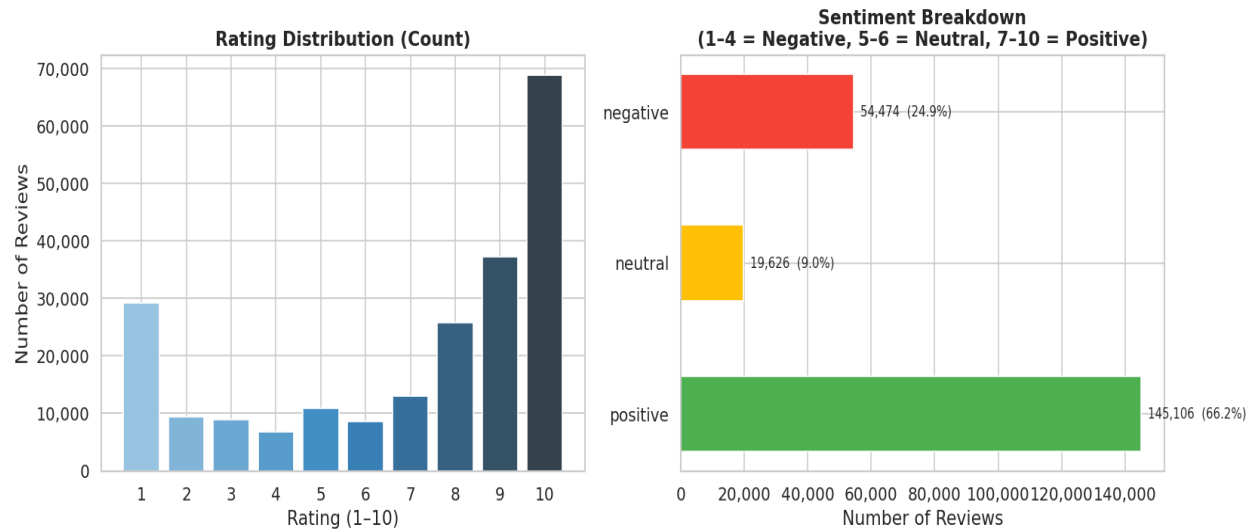


Figure 1: Overall rating distribution and sentiment breakdown.

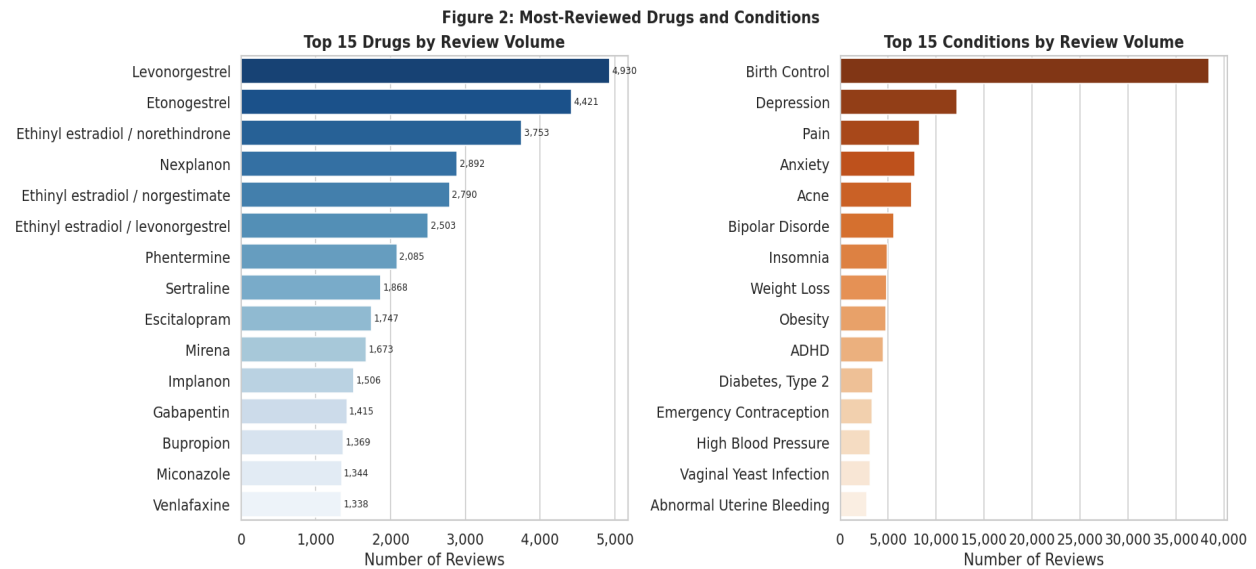


Figure 2: Top 15 most reviewed drugs and conditions.

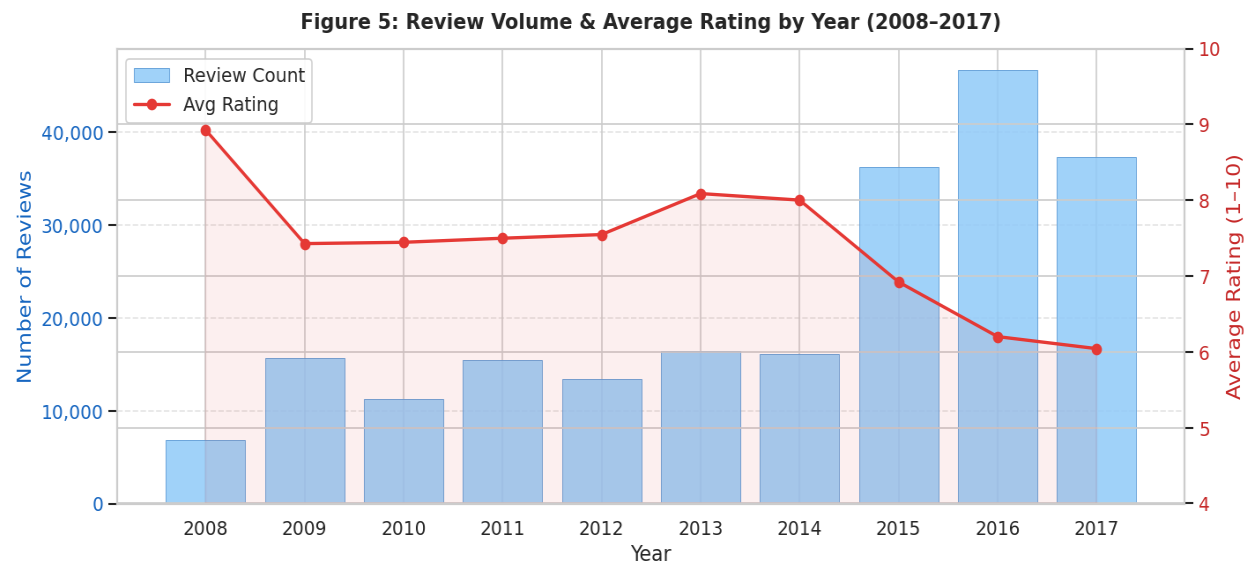


Figure 5: Review volume and average rating by year (2008-2017).

2. Drug Summaries

The following summaries cover the five most reviewed drugs in the dataset. Each summary is generated by the Gemini API based on structured data profiles including review counts, average ratings, sentiment distributions, and representative patient reviews.

Levonorgestrel

Here is a summary report for Levonorgestrel based on the provided patient review data:

Medication Summary Report: Levonorgestrel

1. Patient Experience Overview

Patients submitted a total of 4,930 reviews for Levonorgestrel. The medication received an average rating of 7.38 out of 10. Notably, the median rating was higher at 9.0 out of 10, suggesting a substantial proportion of highly positive experiences despite some lower scores influencing the average.

2. Sentiment Breakdown

A significant majority of reviews, 69.7% (3,435 reviews), were positive. Neutral reviews accounted for 9.9% (488 reviews) of the total, while 20.4% (1,007 reviews) expressed negative sentiment. This indicates that while many patients report satisfactory outcomes, a notable segment experiences less favorable results.

3. Key Themes

Based on the sample patient reviews, common themes include discussions around the medication's effectiveness for its intended purpose, such as contraception, with one patient stating it "worked for me twice." Another recurring theme is the experience of side effects, ranging from "nasty side effects" to "extremely painful and unbearable cramps," which some patients report hindering daily activities. The medication's impact on menstrual periods, including the cessation of painful periods, was also mentioned as a positive initial experience.

4. Clinical Relevance

When counseling patients on Levonorgestrel, pharmacists should be aware that while a large majority of patients (69.7%) report positive experiences, a substantial 20.4% report negative outcomes. Patients commonly use this medication for conditions such as Birth Control, Emergency Contraception, and Abnormal Uterine Bleeding. Discussion points could include the potential for effective relief from painful periods or prevention of pregnancy, as well as the possibility of significant side effects like cramps, which patients have described as severe.

Etonogestrel

Here is a summary report of patient reviews for Etonogestrel:

1. Patient Experience Overview

Etonogestrel has received 4,421 patient reviews, with an average rating of 5.82 out of 10. The median rating stands at 6.0 out of 10. These figures suggest a moderate overall level of patient satisfaction reported in the reviews.

2. Sentiment Breakdown

Patient sentiment is diverse, with positive reviews accounting for 47.9% (2,118 reviews) of the total. Negative experiences were reported by 38.6% (1,706 reviews) of patients, while neutral reviews constituted 13.5% (597 reviews). This indicates a nearly even distribution between positive and negative feedback regarding the medication.

3. Key Themes

Based on the provided sample reviews, recurring themes include changes in menstrual cycles, such as unpredictable bleeding, prolonged cycles, or the cessation of periods. Patients also reported side effects like weight gain, decreased sex drive, nausea, constant headaches, and feelings of confusion or generally "not feeling oneself." The ease of insertion and removal was mentioned positively in one review.

4. Clinical Relevance

Pharmacists counseling patients on Etonogestrel should be aware of the varied patient experiences, with nearly half of reviews being positive (47.9%) and a substantial portion being negative (38.6%). Patient-reported side effects like menstrual cycle irregularities, weight gain, decreased libido, nausea, and headache are common themes. Discussing these potential experiences can help manage patient expectations and identify adverse effects more effectively.

Ethinyl estradiol / norethindrone

Here is a summary report based on the provided patient review data for Ethinyl Estradiol / Norethindrone:

1. Patient Experience Overview

The medication Ethinyl Estradiol / Norethindrone has garnered 3,753 patient reviews. The average patient rating stands at 5.59 out of 10, with a median rating of 6.0 out of 10. These figures suggest a moderate level of overall satisfaction among the patient population who submitted reviews.

2. Sentiment Breakdown

Of the total reviews, 1,771 (47.2%) were positive, indicating a favorable experience for nearly half of the reviewers. Neutral experiences accounted for 435 (11.6%) of the reviews, while 1,547 (41.2%) were negative. This shows a relatively balanced sentiment distribution, with positive reviews slightly outnumbering negative ones.

3. Key Themes

Based on the sample patient reviews, recurring themes include the medication's effectiveness in restoring menstrual cycles for conditions like secondary amenorrhea, with some patients reporting no negative effects on weight. Conversely, some patients reported negative impacts such as a significant decrease in libido and the development or worsening of acne.

4. Clinical Relevance

Pharmacists counseling patients on Ethinyl Estradiol / Norethindrone should note the medication's moderate overall satisfaction (average rating 5.59/10) and the nearly equal distribution of positive (47.2%) and negative (41.2%) experiences. When discussing potential effects, it may be relevant to address common patient-reported themes, including its utility for conditions like Secondary Amenorrhea, potential for weight changes, and side effects such as altered libido or the onset of acne.

Nexplanon

Here is a summary report based on the Nexplanon patient review data:

1. Patient Experience Overview

Based on 2,892 patient reviews, Nexplanon received an average rating of 5.64 out of 10. The median rating for this medication was 6.0 out of 10, indicating a central tendency slightly above the average. These figures suggest a varied overall patient satisfaction level.

2. Sentiment Breakdown

Patient reviews showed a notable distribution of sentiment, with positive experiences comprising 45.6% (1,318 reviews) of the total. Negative experiences were also substantial, accounting for 40.9% (1,184 reviews). Neutral reviews represented 13.5% (390 reviews) of the reported patient feedback.

3. Key Themes

The sample reviews highlight several recurring themes, including the necessity of an initial adjustment period for the body to adapt to the contraception. Patients frequently mentioned experiencing mood swings or emotional changes, and reported issues such as unpredictable or prolonged bleeding patterns. Other concerns noted were weight gain, skin breakouts, and gastrointestinal problems like constipation.

4. Clinical Relevance

Pharmacists should be aware of Nexplanon's mixed patient reception, reflected by an average rating of 5.64/10 and nearly balanced positive (45.6%) and negative (40.9%) sentiment. When counseling patients, it would be beneficial to discuss the potential for an initial adjustment period and common side effects. Key topics for discussion could include mood changes, variations in bleeding patterns, and possible weight fluctuations, which were frequently cited in patient experiences.

Ethinyl estradiol / norgestimate

Here is a summary report for Ethinyl Estradiol / Norgestimate based on the provided patient review data.

1. Patient Experience Overview

Based on 2,790 patient reviews, the overall satisfaction with Ethinyl Estradiol / Norgestimate appears moderate. The medication received an average rating of 5.80 out of 10, with a median rating of 6.0 out of 10. These figures suggest a diverse range of experiences among users, leaning slightly towards the positive midpoint.

2. Sentiment Breakdown

The sentiment breakdown indicates a divided patient experience for this medication. Approximately half of the reviews are classified as positive (1,379 reviews, 49.4%), suggesting satisfaction for a significant portion of users. However, a substantial number of reviews are negative (1,054 reviews, 37.8%), while neutral reviews account for a smaller segment (357 reviews, 12.8%).

3. Key Themes

Sample patient reviews highlight both desired therapeutic effects and notable adverse experiences. Patients mention the medication's effectiveness as birth control and its initial impact on managing painful periods. Conversely, recurring concerns include side effects such as nausea, weight gain (e.g., "gained 8 lbs in a month"), acne, and changes in menstrual bleeding, with some reporting heavier periods and bad cramps. One review also indicated a shift from initial improvement to worsening symptoms after approximately three months.

4. Clinical Relevance

When counseling patients on Ethinyl Estradiol / Norgestimate, pharmacists should acknowledge the diverse patient experiences indicated by the average rating of 5.80/10 and the nearly balanced split between positive (49.4%) and negative (37.8%) reviews. It is relevant to discuss the primary indications, such as birth control, acne, abnormal uterine bleeding, and ovarian cysts, as well as potential side effects like nausea, weight gain, acne, and changes in menstrual bleeding patterns, which are recurring themes in patient feedback. Given some reports of delayed onset of negative effects, pharmacists might also consider discussing what to expect during the initial months of therapy.

3. Condition Summaries

The following summaries cover the five most reviewed medical conditions in the dataset. Each summary includes a condition overview, drug comparison, patient sentiment themes, and research implications generated by the Gemini API.

Birth Control

Here is a summary report for the BIRTH CONTROL condition based on the provided patient review data:

1. Condition Overview

The BIRTH CONTROL condition has accumulated a substantial volume of 38,504 patient reviews. The overall treatment satisfaction for this condition averages 6.08 out of 10, with a median rating of 7.0 out of 10. A majority of reviews, 52.3% (20,122 reviews), are positive, while 35.1% (13,520 reviews) are negative, and 12.6% (4,862 reviews) are neutral.

2. Drug Comparison

Among the top five most reviewed drugs for Birth Control, Etonogestrel has the highest review count with 4,394 reviews, averaging 5.8 out of 10. Levonorgestrel stands out with the highest average rating of 7.0 out of 10 from 2,884 reviews. Other top drugs include Ethinyl estradiol / norethindrone (3,081 reviews, avg 5.6), Nexplanon (2,883 reviews, avg 5.6), and Ethinyl estradiol / levonorgestrel (2,107 reviews, avg 5.9).

3. Patient Sentiment Themes

Based on the provided sample reviews, positive experiences highlight benefits such as lighter periods, as indicated by a 9/10 rating for Ortho Cyclen. Negative feedback, exemplified by a 4/10 rating for Sronyx, points to concerns like weight gain (e.g., about 8 lbs). A neutral review for Ethinyl estradiol / norgestimate (rated 5/10) describes issues such as worsened PMS symptoms, bloating, tender breasts, exhaustion, anxiety, and decreased sex drive, despite initial benefits.

4. Research Implications

Further investigation could explore the factors contributing to the difference between the average rating (6.08) and the median rating (7.0), which might suggest a skew in patient satisfaction distribution. The notable variation in average ratings among the top medications, ranging from 5.6 to 7.0, warrants deeper analysis into the specific characteristics or patient populations associated with higher satisfaction. Additionally, examining the prevalence and impact of adverse effects like weight gain and changes in mood or libido, as mentioned in the sample reviews, could offer valuable insights.

Depression

Here is a summary report for the DEPRESSION condition profile:

1. Condition Overview

The condition DEPRESSION has accumulated a substantial volume of 12,466 patient reviews. Overall patient satisfaction is notably positive, with an average rating of 7.09 out of 10 and a higher median rating of 8.0 out of 10. The majority of reviews are positive, accounting for 68.6% (8555 reviews), while 22.1% (2758 reviews) are negative, and 9.2% (1153 reviews) are neutral.

2. Drug Comparison

Among the most frequently reviewed medications for DEPRESSION, Bupropion received the highest number of reviews with 747, and also achieved a strong average rating of 7.4. Desvenlafaxine and Pristiq closely followed in average rating at

7.2, from 573 and 554 reviews respectively. Venlafaxine, with 574 reviews, recorded the lowest average rating among these top five drugs, at 6.5.

3. Patient Sentiment Themes

Positive patient sentiments often describe a significant improvement in mood and a renewed perspective, such as experiencing "inner peace" and feeling "content" shortly after starting medication. In contrast, negative sentiments frequently report unbearable side effects, including worsening mood, physical symptoms like vomiting, and increased suicidal ideation. Neutral reviews typically express a lack of significant change in condition ("still feel blah") or challenges in managing side effects like drowsiness.

4. Research Implications

The difference between the average rating (7.09) and the median rating (8.0) suggests that a subset of very low ratings may be disproportionately impacting the overall average, warranting further investigation into these specific experiences. Further analysis could explore the factors behind the varied average ratings among the top-reviewed drugs, particularly identifying contributors to Venlafaxine's lower average rating of 6.5 compared to Bupropion's 7.4. Examining the complete review data for common side effects or time to efficacy across these medications could provide more granular insights.

Pain

Here is a summary report for the PAIN condition profile:

1. Condition Overview

The 'PAIN' condition profile includes 8,259 patient reviews, indicating a significant volume of patient feedback. Overall treatment satisfaction appears moderately high, with an average rating of 7.62 out of 10 and a median rating of 9.0 out of 10. The majority of reviews are positive, accounting for 76.5% (6,319 reviews), while 17.2% (1,423 reviews) are negative and 6.3% (517 reviews) are neutral.

2. Drug Comparison

Among the top five drugs for this condition, Tramadol received the highest number of reviews at 501, but registered a lower average rating of 6.5 out of 10. Oxycodone had a high average rating of 8.8 across 380 reviews, closely followed by Acetaminophen / oxycodone with an average rating of 8.7 from 363 reviews. Acetaminophen / hydrocodone also showed a strong average rating of 8.0 from 452 reviews, while Tapentadol, with 285 reviews, also had an average rating of 6.5.

3. Patient Sentiment Themes

Positive sentiment themes, as observed in the sample, include perceived efficacy and value, with one reviewer noting a drug "worked better" and was "cheaper." Conversely, negative sentiment is directly linked to a lack of effectiveness, exemplified by comments such as "Does nothing may as well take sugar pills." Neutral reviews highlight concerns about side effects, such as constant tiredness, and perceived insufficient pain management, alongside comparisons to previous medications.

4. Research Implications

The notable difference between the average rating of 7.62 and the median rating of 9.0 for PAIN suggests a potential skewing by lower scores or outliers, warranting further investigation into the distribution of ratings. Further research could explore the specific factors contributing to the lower average ratings for drugs like Tramadol and Tapentadol (both 6.5) compared to other top-reviewed medications. Additionally, a deeper qualitative analysis of neutral and negative reviews could reveal common patterns in patient-reported side effects or efficacy concerns that lead to dissatisfaction.

Anxiety

Here is a summary report for the ANXIETY condition based on the provided data:

1. Condition Overview

For the condition Anxiety, 7,902 patient reviews were analyzed. The overall average rating is 7.68 out of 10, with a median rating of 9.0 out of 10. A significant majority of reviews, 76.4% (6040 reviews), are positive, while 17.3% (1370 reviews) are negative and 6.2% (492 reviews) are neutral.

2. Drug Comparison

Among the top reviewed medications for Anxiety, Escitalopram has the highest volume with 669 reviews and an average rating of 7.8. Alprazolam received 560 reviews with a high average rating of 8.8, while Clonazepam also showed a strong average rating of 8.6 from 528 reviews. In contrast, Buspirone had 538 reviews but a lower average rating of 6.3.

3. Patient Sentiment Themes

Positive patient reviews, such as for Lexapro, frequently describe the medication as a "life saver" and highlight its effectiveness in preventing anxiety and panic attacks, despite potential side effects like reduced sex drive. Conversely, negative and neutral reviews, exemplified by those for BuSpar, often report experiencing new or worsening symptoms like dizziness, fatigue, headaches, chest pressure, and muscle twitches. Some neutral reviews indicate partial promise but still necessitate additional medication for symptom management.

4. Research Implications

The substantial difference between the average rating (7.68) and the median rating (9.0) suggests a skewed distribution of patient satisfaction, warranting further investigation into the specific factors driving this pattern. The wide range of average ratings among top-reviewed drugs, from 8.8 for Alprazolam to 6.3 for Buspirone, indicates significant variability in patient experiences with different medications. Further research could explore the reasons behind these divergent outcomes, including specific side effect profiles or patient demographics associated with distinct satisfaction levels.

Acne

Here is a condition summary report for ACNE based on the provided data:

1. Condition Overview

The "ACNE" condition has a substantial volume of patient feedback, with 7,646 total reviews. Overall treatment satisfaction leans positive, with 71.5% (5,464 reviews) classified as positive. The average rating is 7.36 out of 10, while the median rating is notably higher at 9.0 out of 10, indicating a skew towards higher satisfaction among a majority of reviewers despite 20.0% (1,532 reviews) being negative.

2. Drug Comparison

Among the top drugs for acne, Isotretinoin (682 reviews) and Accutane (431 reviews) received the highest average ratings, both at 8.4. Adapalene / benzoyl peroxide (547 reviews) and Epiduo (483 reviews) shared an average rating of 7.4. Doxycycline, with 500 reviews, had an average rating of 7.3.

3. Patient Sentiment Themes

Positive reviews often highlight a treatment's effectiveness, even if initial side effects like burning or tingling occur but eventually subside, as described for Azelex. In contrast, negative sentiments frequently detail severe adverse skin reactions, including extreme dryness, itchiness, patchiness, or a "burned" sensation, as reported for Benzoyl peroxide / clindamycin. Neutral reviews, such as the sample for Isotretinoin, describe a mixed experience involving initial mild side effects, followed by more significant issues like frequent nosebleeds, dry eyes, or an initial worsening of the condition.

4. Research Implications

The notable difference between the average rating (7.36) and median rating (9.0) suggests a deeper investigation into the distribution of patient satisfaction scores could be beneficial. Further analysis of the highly rated drugs, Isotretinoin and Accutane (both avg 8.4), particularly concerning the trade-offs patients accept for perceived efficacy given reported side effects, is warranted. Additionally, the prevalence of severe skin reactions in negative reviews points to a need for research into patient education on side effect management or identifying subgroups more susceptible to these adverse effects.

4. Sentiment & Text Analysis

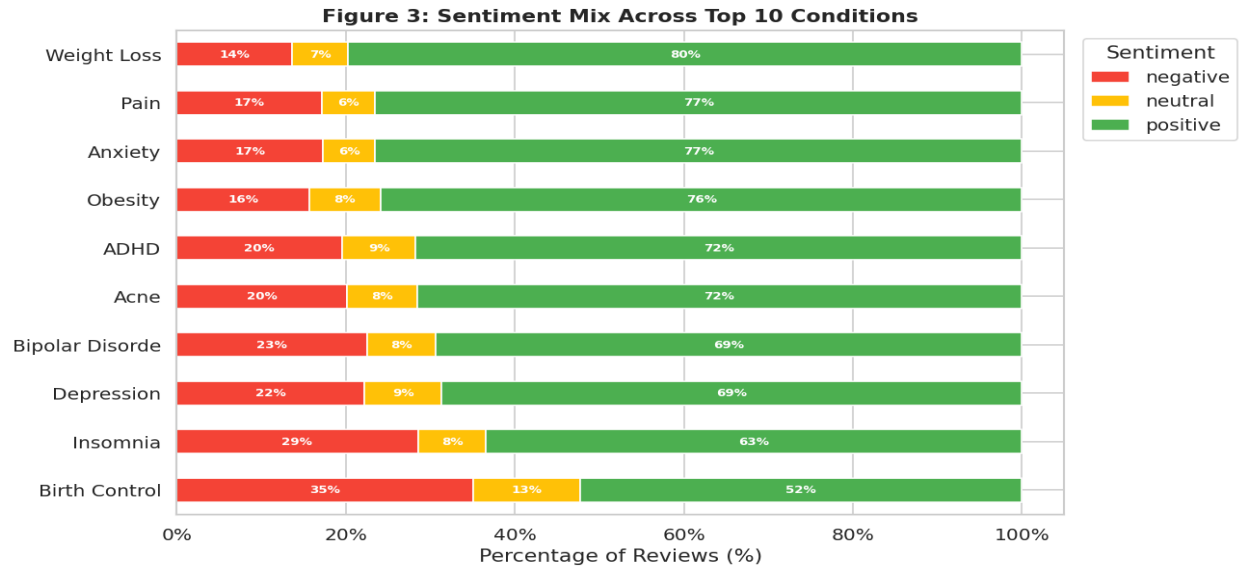


Figure 3: Sentiment mix across the top 10 conditions.

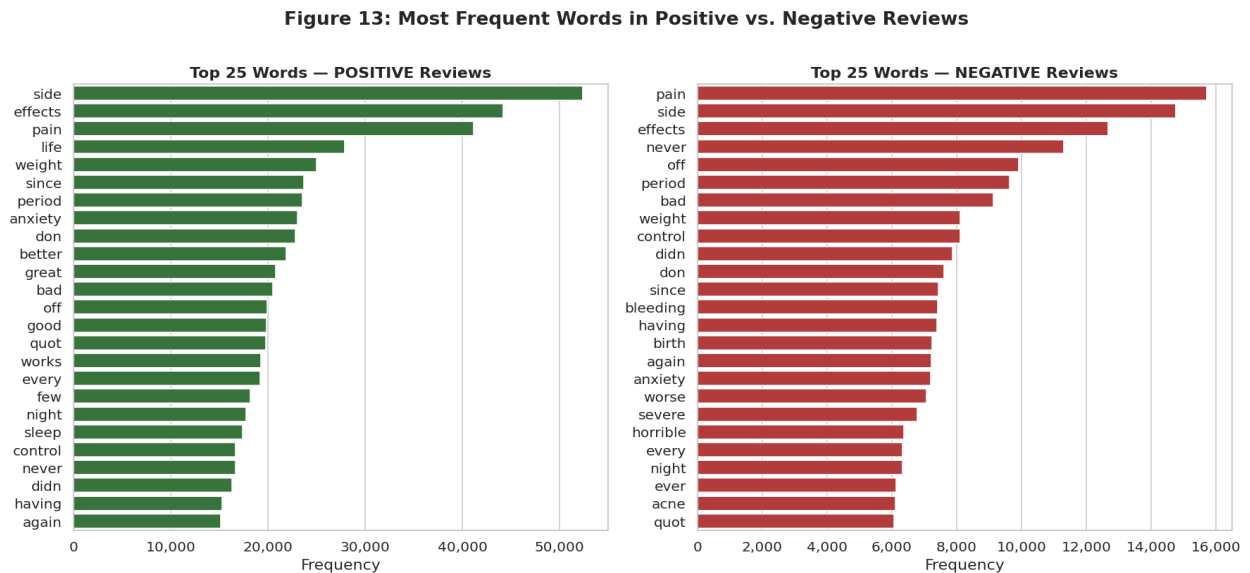


Figure 13: Most frequent words in positive vs. negative reviews.

Figure 14: Word Clouds – Positive vs. Negative Reviews

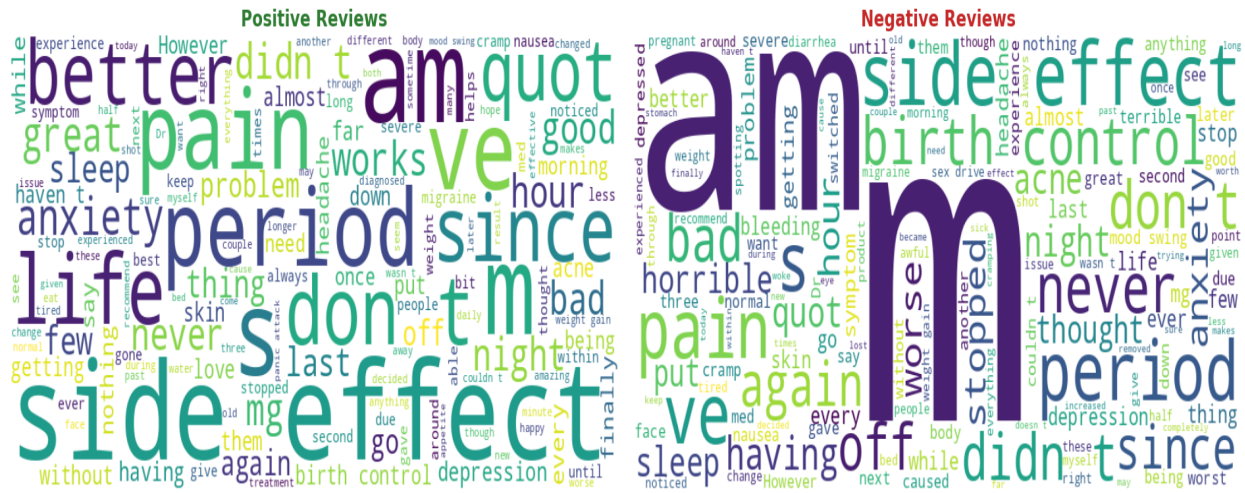


Figure 14: Word clouds for positive and negative reviews.